

# 1 Recommended configuration

Below are the recommended steps for activating a webhook service.

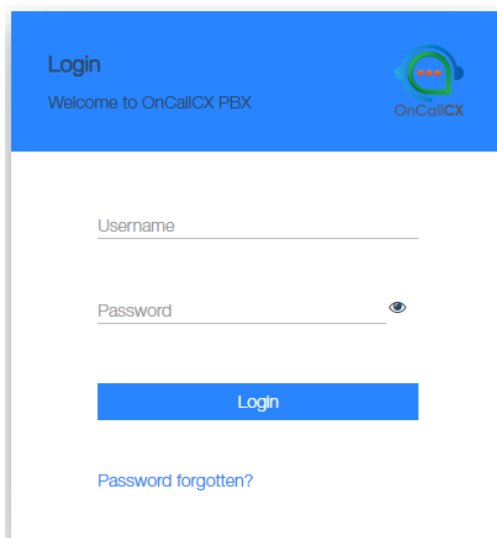
| Step: | Step Description:                                          | Remarks: |
|-------|------------------------------------------------------------|----------|
| 1     | Assign user roles and user attribute to webhook admin user |          |
| 2     | Activate webhook events through webhook config portal      |          |
| 3     |                                                            |          |
| 4     |                                                            |          |

## 1.1 Portal configuration

Below is the procedure to activate.

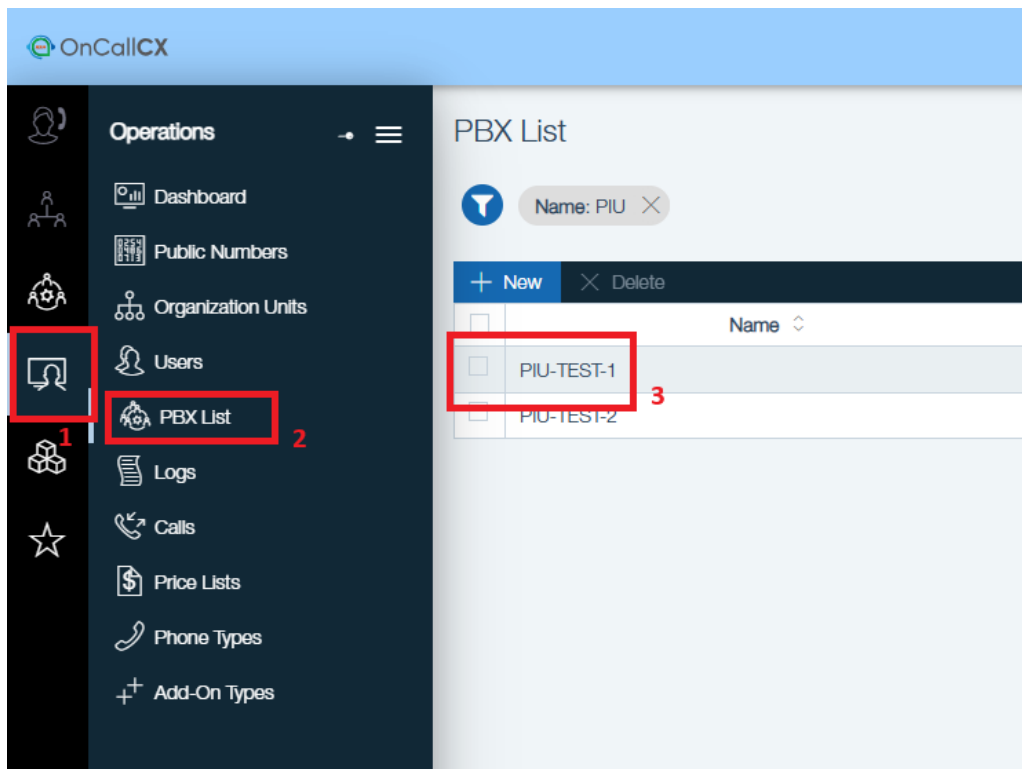
| Step: | Step Description:                                          | Remarks: |
|-------|------------------------------------------------------------|----------|
| 1     | Login to Portal with the Administrator privilege           |          |
| 2     | Determine CTI Device ID                                    |          |
| 3     | Assign user roles and user attribute to webhook admin user |          |
| 4     |                                                            |          |

**Step 1:** Login to OnCallCX PBX Portal with the Administrator privilege

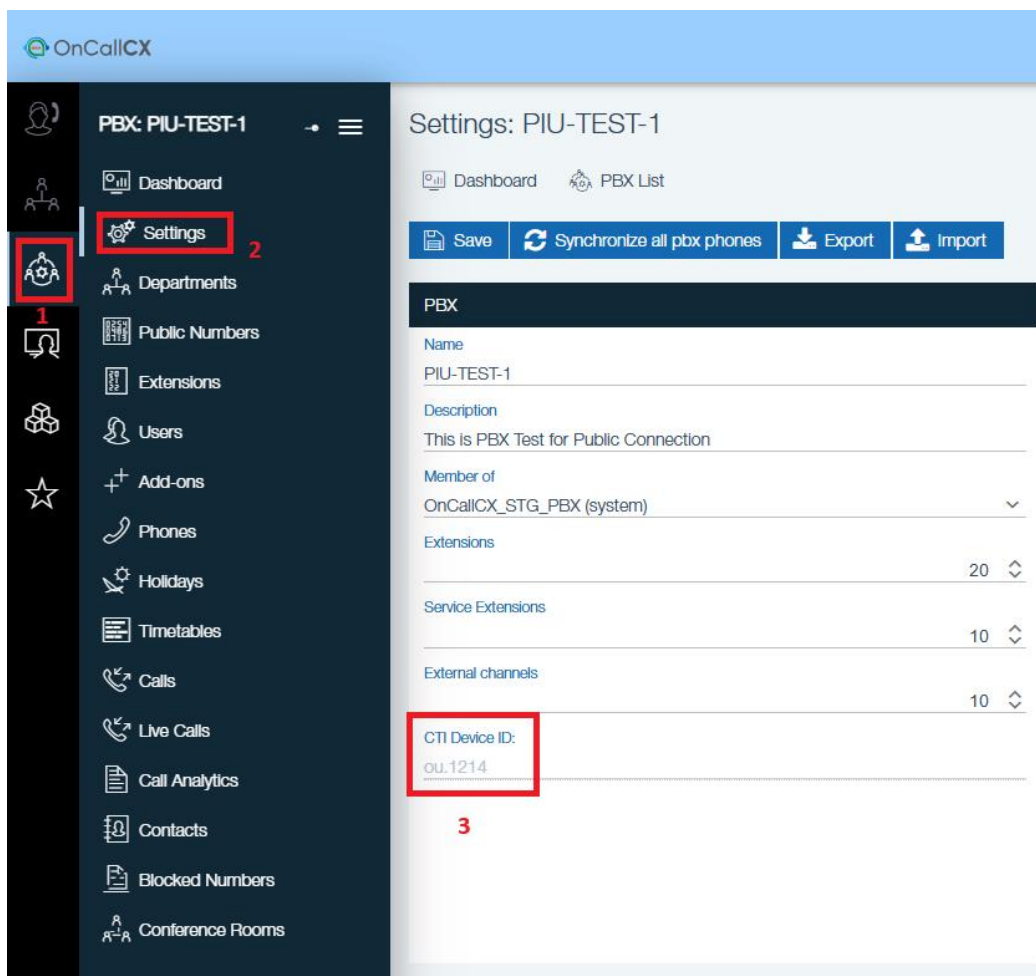


**Step 2:** Determine CTI Device ID

Navigate to **Operations** → **PBX list**, select the PBX need to enable webhook event (e.g: PIU-TEST-1)



To get the CTI Device ID, navigate to **Settings**

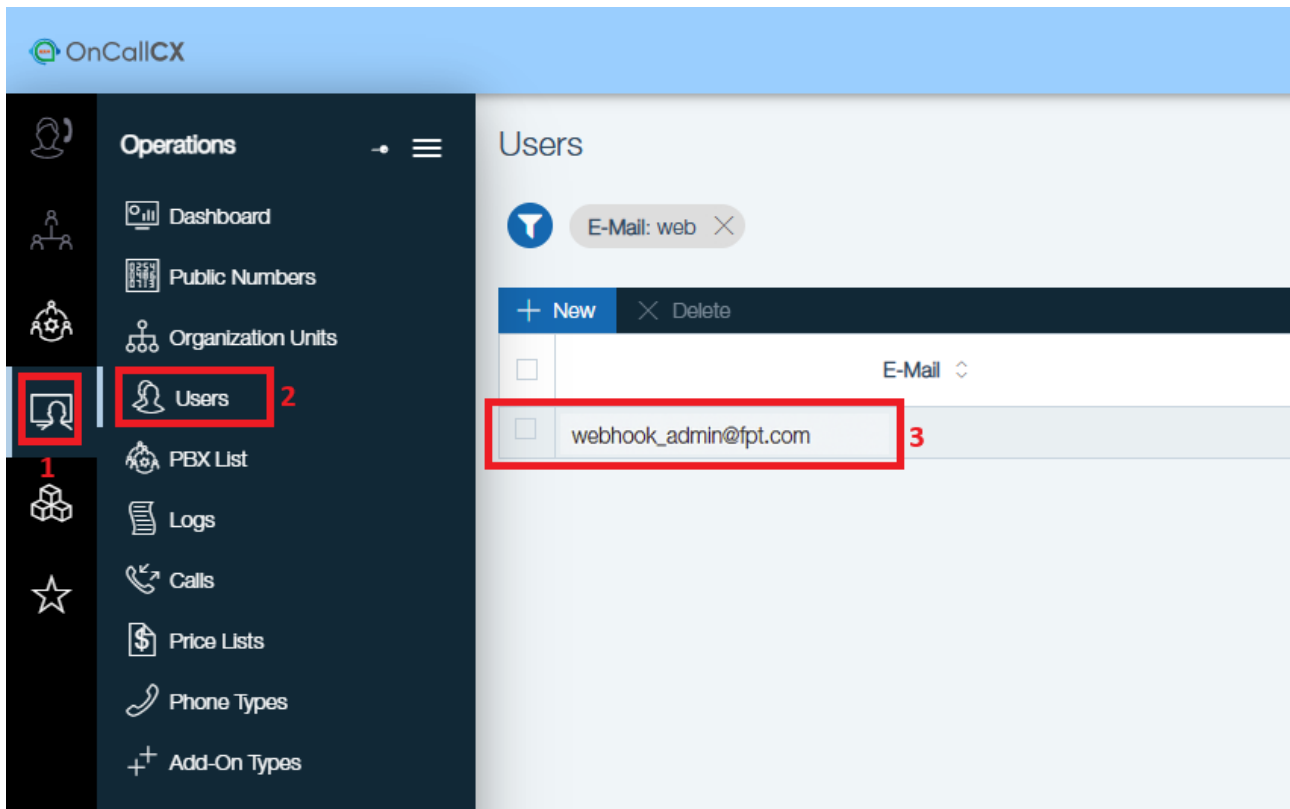


**Step 2:** Assign user roles and user attribute to webhook admin user

**Note**

All the PBXs which enable webhook events are configured under the same webhook admin user

Navigate to **Operations** → **Users**, select the webhook admin user. (e.g: webhook\_admin@fpt.com)



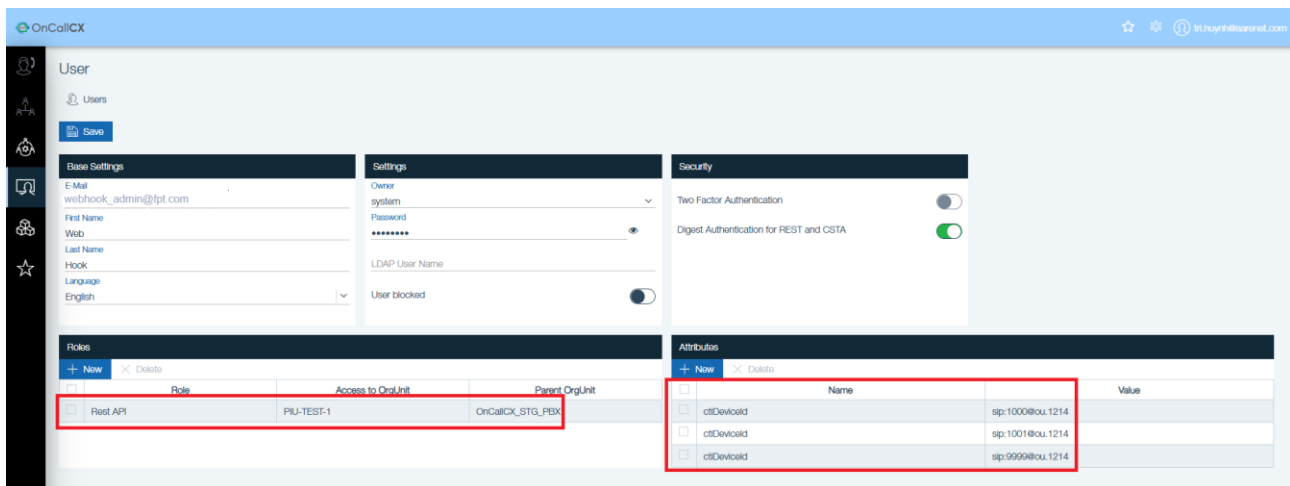
**Create new role and attributes:**

**New Role**

1. **Role:** REST API
2. **OrgUnit:** PBX Name

**New Attributes:**

1. **Name:** ctiDeviceId
2. **Value:** sip:<Extension-Num>@ou.<CTI Device ID>



## 1.2 Webhook configuration

Below is the procedure to activate.

| Step: | Step Description:                                               | Remarks: |
|-------|-----------------------------------------------------------------|----------|
| 1     | Login to Webhook config portal with the Administrator privilege |          |
| 2     | Activate webhook event for PBX                                  |          |
|       |                                                                 |          |

### Step 1: Login to Webhook Config portal

Use OnCallCX PBX account which has Administrator privileges to access to webhook page at: <https://pbx-stg.ocalcx.vn:8892/login>

### Login

Username

Password

Login

## Step 2: Activate webhook for a PBX

After logging in successfully, the portal will let you configure the following:

### App

|     |                                   |
|-----|-----------------------------------|
| URL | https://pbx-stg.ontcallcx.vn:8893 |
|-----|-----------------------------------|

### Webhooks

|               |                                                                     |             |                   |
|---------------|---------------------------------------------------------------------|-------------|-------------------|
| CTI Device ID | Get this value from PBX settings, e.g: ou.2                         | CDR Webhook | Extension Webhook |
| CRM URL       | The URL of customer's CRM, e.g. http://192.168.1.10:8890/test/event |             |                   |

☒ Active (checked = true, unchecked = false)

**App URL:** the URL of webhook server: https://<pbx-ontcallcx-fqdn>:8893, e.g.: https://pbx-stg.ontcallcx.vn:8893

**CTI Device ID:** This id can be retrieved in PBX settings, e.g.: ou.2

**CRM URL:** The customer's CRM URL which wants to receive webhook events from OntcallCX PBX, e.g.: http://customer-crm.company.com/events

**Active:** Enable or disable webhook events

### Button:

- CDR Webhook: allow OntcallCX PBX to send CDR events to the customer.
- Extension Webhook: allow OntcallCX PBX to send extension events to the customer.

## 2 Event Reference

### 2.1 CDR Events

#### call\_start

| Key name   | Value                                                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| event_type | call_start                                                                                                                        |
| call_id    | The ID of call                                                                                                                    |
| session_id | The ID of call                                                                                                                    |
| callee     | The dialed number                                                                                                                 |
| caller     | The dialer number                                                                                                                 |
| cdr_id     | The ID of call                                                                                                                    |
| direction  | Classification of incoming/outgoing calls.<br>There are 2 possible value: <b>in</b> – incoming call<br><b>out</b> – outgoing call |
| tenant_id  | The ID of tenant                                                                                                                  |
| time       | Event time                                                                                                                        |

#### Example:

```
{  
  "event_type": "call_start",  
  "call_id": "cc1250918152610380",  
  "session_id": "cc1250918152610380",  
  "callee": "*14",  
  "caller": "1000",  
  "cdr_id": "cc1250918152610380",  
  "direction": "out",  
  "tenant_id": "ou.2",  
  "Time": "2025-09-18T08:26:09.221Z"  
}
```

#### cdr\_target\_ringing

| Key name    | Value                                                            |
|-------------|------------------------------------------------------------------|
| event_type  | cdr_target_ringing                                               |
| caller      | The dialer number                                                |
| session_id  | The ID of call                                                   |
| destination | Information about the number being called is currently connected |

|           |                                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| cdr_id    | The ID of call                                                                                                                    |
| direction | Classification of incoming/outgoing calls.<br>There are 2 possible value: <b>in</b> – incoming call<br><b>out</b> – outgoing call |
| did       | Hotline number to receive incoming calls                                                                                          |
| tenant_id | The ID of tenant                                                                                                                  |
| time      | Event time                                                                                                                        |

**Example:**

```
{
  "event_type": "cdr_target_ringing",
  "Caller": "1000",
  "session_id": "cc1250918152610380",
  "Destination": "*14",
  "cdr_id": "cc1250918152610380",
  "direction": "out",
  "tenant_id": "ou.2",
  "Time": "2025-09-18T08:26:09.236Z"
}
```

### cdr\_target\_answer

| Key name    | Value                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| event_type  | cdr_target_answer                                                                                                                 |
| caller      | The dialer number                                                                                                                 |
| session_id  | The ID of call                                                                                                                    |
| destination | Information about the number being called is currently connected                                                                  |
| cdr_id      | The ID of call                                                                                                                    |
| direction   | Classification of incoming/outgoing calls.<br>There are 2 possible value: <b>in</b> – incoming call<br><b>out</b> – outgoing call |
| did         | Hotline number to receive incoming calls                                                                                          |
| tenant_id   | The ID of tenant                                                                                                                  |
| time        | Event time                                                                                                                        |

### Example:

```
{  
  "event_type": "cdr_target_answer",  
  "Caller": "1000",  
  "session_id": "cc1250918152610380",  
  "Destination": "*14",  
  "cdr_id": "cc1250918152610380",  
  "direction": "out",  
  "tenant_id": "ou.2",  
  "Time": "2025-09-18T08:26:09.240Z"  
}
```

### cdr\_target\_noanswer

| Key name    | Value                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| event_type  | cdr_target_noanswer                                                                                                               |
| caller      | The dialer number                                                                                                                 |
| session_id  | The ID of call                                                                                                                    |
| destination | Information about the number being called is currently connected                                                                  |
| cdr_id      | The ID of call                                                                                                                    |
| direction   | Classification of incoming/outgoing calls.<br>There are 2 possible value: <b>in</b> – incoming call<br><b>out</b> – outgoing call |
| did         | Hotline number to receive incoming calls                                                                                          |



|           |                  |
|-----------|------------------|
| tenant_id | The ID of tenant |
| time      | Event time       |

**Example:**

```
{
  "event_type": "cdr_target_noanswer",
  "Caller": "1000",
  "session_id": "cc1250918160148387",
  "Destination": "02873030153",
  "cdr_id": "cc1250918160148387",
  "direction": "out",
  "tenant_id": "ou.2",
  "Time": "2025-09-18T09:01:47.823Z"
}
```

**cdr\_target\_end**

| Key name    | Value                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| event_type  | cdr_target_end                                                                                                                    |
| caller      | The dialer number                                                                                                                 |
| session_id  | The ID of call                                                                                                                    |
| destination | Information about the number being called is currently connected                                                                  |
| cdr_id      | The ID of call                                                                                                                    |
| direction   | Classification of incoming/outgoing calls.<br>There are 2 possible value: <b>in</b> – incoming call<br><b>out</b> – outgoing call |
| did         | Hotline number to receive incoming calls                                                                                          |
| tenant_id   | The ID of tenant                                                                                                                  |
| time        | Event time                                                                                                                        |

**Example:**

```
{
  "event_type": "cdr_target_end",
  "Caller": "1000",
  "session_id": "cc1250918152610380",
  "Destination": "*14",
  "cdr_id": "cc1250918152610380",
  "direction": "out",
  "tenant_id": "ou.2",
}
```

```
"Time": "2025-09-18T08:26:15.022Z"
}
```

#### cdr\_target\_fail

| Key name    | Value                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| event_type  | cdr_target_fail                                                                                                                   |
| caller      | The dialer number                                                                                                                 |
| session_id  | The ID of call                                                                                                                    |
| destination | Information about the number being called is currently connected                                                                  |
| cdr_id      | The ID of call                                                                                                                    |
| direction   | Classification of incoming/outgoing calls.<br>There are 2 possible value: <b>in</b> – incoming call<br><b>out</b> – outgoing call |
| did         | Hotline number to receive incoming calls                                                                                          |
| tenant_id   | The ID of tenant                                                                                                                  |
| time        | Event time                                                                                                                        |

#### Example:

```
{
  "event_type": "cdr_target_fail",
  "Caller": "1000",
  "session_id": "cc1251004153557142",
  "Destination": "2000",
  "cdr_id": "cc1251004153557142",
  "direction": "out",
  "tenant_id": "ou.2",
  "Time": "2025-10-04T08:35:57.526Z"
}
```

#### call\_cdr

| Key name      | Value                          |
|---------------|--------------------------------|
| event_type    | call_cdr                       |
| call_id       | The ID of call                 |
| session_id    | The ID of call                 |
| callee        | The dialed number              |
| caller        | The dialer number              |
| tenant_id     | The ID of tenant               |
| answered_time | Timestamp the call is answered |

|              |                                                                                                                                                                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| start_time   | Timestamp the call is started                                                                                                                                                                                            |
| ring_time    | Timestamp the call is ringing                                                                                                                                                                                            |
| ended_time   | Timestamp the call is ended                                                                                                                                                                                              |
| direction    | Classification of incoming/outgoing calls.<br>There are 2 possible value: <b>in</b> – incoming call<br><b>out</b> – outgoing call                                                                                        |
| end_reason   | The call end reason, possible values:<br>invalidNumberFormat<br>blocked<br>numberUnallocated<br>timeout<br>destNotObtainable<br>busy<br>normalClearing<br>conference<br>transfer<br>redirected<br>callPickup<br>callBack |
| did_cid      | Hotline number to receive incoming calls                                                                                                                                                                                 |
| call_targets | {<br>answered_time<br>caller<br>callee<br>destination<br>start_time<br>ring_time<br>end_time<br>}                                                                                                                        |

### Examples:

```
{
  "event_type": "call_cdr",
  "call_id": "cc22510311054396816",
  "session_id": "cc22510311054396816",
  "callee": "8888",
  "caller": "1001",
  "tenant_id": "ou.1214",
  "answered_time": "2025-10-31T03:54:40.216Z",
```

```
"start_time": "2025-10-31T03:54:39.630Z",
"ring_time": "2025-10-31T03:54:39.633Z",
"ended_time": "2025-10-31T03:54:41.512Z",
"direction": "out",
"end_reason": "normalClearing",
"call_targets": [
  {
    "answered_time": "2025-10-31T03:54:40.216Z",
    "caller": "1001",
    "callee": "8888",
    "destination": "8888",
    "id": "cc22510311054396816",
    "start_time": "2025-10-31T03:54:39.630Z",
    "ring_time": "2025-10-31T03:54:39.633Z",
    "end_time": "2025-10-31T03:54:41.512Z"
  }
]
```

## 2.2 DTMF Events

### dtmf\_event

| Key name     | Value                                        |
|--------------|----------------------------------------------|
| event_type   | dtmf_event                                   |
| call_id      | The ID of call                               |
| extension    | The internal number                          |
| extension_id | The ID of internal number                    |
| dtmfChar     | The DTMF characters: 0,1,2,3,4,5,6,7,8,9,#,* |
| tenant_id    | The ID of tenant                             |
| time         | Event time                                   |

### Example:

```
{
  "event_type": "dtmf_event",
  "call_id": "cc1250920224900128",
  "Extension": "1000",
  "extension_id": "sip:1000@ou.2",
  "dtmfChar": "5",
  "Time": "2025-09-20T15:49:01.422Z",
  "tenant_id": "ou.2"
}
```

## 2.3 Extension Events

### extension\_status

| Key name      | Value                                                                                                                                                                                                                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| event_type    | extension_status                                                                                                                                                                                                                                                                                                |
| call_status   | Always ""                                                                                                                                                                                                                                                                                                       |
| extension     | The internal number                                                                                                                                                                                                                                                                                             |
| extension_id  | The ID of internal number                                                                                                                                                                                                                                                                                       |
| online        | The registration status of extension:<br><b>true</b> : register<br><b>false</b> : un-register                                                                                                                                                                                                                   |
| presence_note | The presence state of extension:<br><br>/* registration state */<br>offline?: boolean;<br><br>/* presence-states */<br>available?: boolean;<br>busy?: boolean;<br>dnd?: boolean;<br>away?: boolean;<br><br>/* substates of busy (only set together with 'busy') */<br>holding?: boolean;<br>alerting?: boolean; |
| tenant_id     | The ID of tenant                                                                                                                                                                                                                                                                                                |
| time          | Event time                                                                                                                                                                                                                                                                                                      |